Excel Dates for Data Analysts

A practical guide with worked examples and data tables

Dates are the backbone of almost every analysis: sales trends, cohort reports, SLA tracking, forecasting. Yet they are also the most common source of silent errors in Excel. This guide shows how Excel really handles dates, the functions you will use every day, and a complete worked example you can rebuild in minutes.

What is inside

- How Excel stores dates and times
- Formatting dates (and why formatting is not conversion)
- The core date functions, with results
- Date differences, working days and month-end logic
- Building a date (calendar) table
- Worked example: order data, SLA and monthly summary
- Cleaning text dates, common errors and fixes
- Best practices and practice exercises with answers

1. How Excel stores dates

Excel does not store a date as text. It stores a **serial number**: the count of days since a starting point. The time of day is stored as the **decimal part** of that number. Because dates are numbers, you can add, subtract, compare and average them.

Table 1. Dates and their serial numbers (Windows 1900 date system)

Date	Serial number	Note
01-Jan-2000	`36526`	Start of the century
01-Jan-2020	`43831`	Start of 2020
01-Jan-2026	`46023`	Start of this year
15-Oct-2026	`46310`	A Thursday

Table 2. Time is a fraction of a day

Time	Stored value	Formula that builds it
06:00	`0.25`	`=TIME(6,0,0)`
12:00	`0.5`	`=TIME(12,0,0)`
18:00	`0.75`	`=TIME(18,0,0)`
14:30	`0.604167`	`=TIME(14,30,0)`
15-Oct-2026 14:30	`46310.604167`	`=DATE(2026,10,15)+TIME(14,30,0)`

> **Tip** To see the serial number behind any date, change the cell format to **General**. If a date ever displays as a number like 46310, the value is fine; only the format is missing.

Good to know Excel treats 1900 as a leap year (it was not) for compatibility with old software, so serials before 1-Mar-1900 are off by one. This never matters for modern data. Excel for Mac historically used a 1904 date system; check **File > Options > Advanced** if dates from another workbook are exactly 4 years and 1 day off.

2. Formatting dates

Formatting changes only what you **see**, never the underlying value. Press `Ctrl+1`, choose **Custom**, and type a format code. The table below shows one value, 15-Oct-2026 14:30, in different formats.

Table 3. Common custom date formats

Format code	Displays	Typical use
`dd/mm/yyyy`	15/10/2026	Day-first reports
`mm/dd/yyyy`	10/15/2026	Month-first reports
`yyyy-mm-dd`	2026-10-15	ISO format; safest for sharing and sorting
`d mmm yyyy`	15 Oct 2026	Readable, unambiguous
`ddd`	Thu	Weekday labels on charts
`dddd`	Thursday	Full weekday name
`mmm-yy`	Oct-26	Monthly axis labels
`mmmm yyyy`	October 2026	Report headings
`dd-mmm-yyyy hh:mm`	15-Oct-2026 14:30	Timestamps
`[h]:mm`	38:15 (for 1.59375 days)	Durations that can exceed 24 hours

Watch out `TEXT(A2,"mmm-yy")` returns **text**, not a date. It is great for labels, but you cannot do date maths on it. Keep the real date in one column and the label in another.

3. Core date functions

In every example below, cell **A2** contains **15-Oct-2026**.

Table 4. The functions you will use daily

Function	Formula	Result	What it does
TODAY	`=TODAY()`	Current date	Updates every time the workbook recalculates
NOW	`=NOW()`	Date and time	Current date plus time
DATE	`=DATE(2026,10,15)`	15-Oct-2026	Builds a date from year, month, day
YEAR	`=YEAR(A2)`	2026	Extracts the year
MONTH	`=MONTH(A2)`	10	Extracts the month number
DAY	`=DAY(A2)`	15	Extracts the day of the month
WEEKDAY	`=WEEKDAY(A2)`	5	Sunday = 1 ... Saturday = 7
WEEKDAY	`=WEEKDAY(A2,2)`	4	Monday = 1 ... Sunday = 7
WEEKNUM	`=WEEKNUM(A2)`	42	Week of year; week 1 contains 1 January
ISOWEEKNUM	`=ISOWEEKNUM(A2)`	42	ISO 8601 week number
EDATE	`=EDATE(A2,3)`	15-Jan-2027	Same day, 3 months later
EOMONTH	`=EOMONTH(A2,0)`	31-Oct-2026	Last day of this month
EOMONTH	`=EOMONTH(A2,-1)+1`	01-Oct-2026	First day of this month
TEXT	`=TEXT(A2,"dddd")`	Thursday	Date as formatted text

Tip `TODAY()` and `NOW()` are volatile: they change every day. For a report that must stay reproducible, type the report date in a single input cell and point every formula at it.

4. Date differences, working days and month ends

4.1 Days, months and years between two dates

Subtracting dates gives days. For complete years or months, use `DATEDIF`. Here **A2 = 10-Mar-2024** (start) and **B2 = 15-Oct-2026** (end).

Table 5. Measuring the gap between two dates

Formula	Result	Meaning
`=B2-A2`	949	Total days between the dates
`=DATEDIF(A2,B2,"Y")`	2	Complete years
`=DATEDIF(A2,B2,"M")`	31	Complete months
`=DATEDIF(A2,B2,"YM")`	7	Months left after removing whole years
`=DATEDIF(A2,B2,"MD")`	5	Days left after removing whole months
`=NETWORKDAYS(A2,B2)`	679	Working days (Mon to Fri), both ends included

Put the three `DATEDIF` pieces together for a readable age or tenure: **2 years, 7 months, 5 days**.

```
=DATEDIF(A2,B2,"Y")&" years, "&DATEDIF(A2,B2,"YM")&" months, "&DATEDIF(A2,B2,"MD")&" days"
```

Note `DATEDIF` is an older function that does not appear in Excel's autocomplete, but it works. The start date must not be later than the end date, otherwise you get `#NUM!`.

4.2 Working days

Table 6. Working-day calculations

Formula	Result	Meaning
`=NETWORKDAYS(DATE(2026,10,1),DATE(2026,10,31))`	22	Working days in October 2026 (Sat/Sun off)
`=WORKDAY(DATE(2026,10,15),10)`	29-Oct-2026	10 working days after 15-Oct-2026
`=NETWORKDAYS.INTL(DATE(2026,10,1),DATE(2026,10,31),7)`	21	Same month with a Friday/Saturday weekend (code 7)

All three functions accept an optional holidays range as the last argument, for example `=NETWORKDAYS(A2,B2,Holidays)`. The `.INTL` versions let you choose any weekend pattern.

4.3 Month-end and fiscal logic

Rolling dates by months is safer with `EDATE` and `EOMONTH` than by adding 30 days, because month lengths vary. Fiscal calendars are just arithmetic on the month number. Assuming a fiscal year that starts in April and is named by the year it ends:

Table 7. Calendar versus fiscal periods for A2 = 15-Oct-2026

Measure	Formula	Result
Calendar quarter	`="Q"&ROUNDUP(MONTH(A2)/3,0)`	Q4
Fiscal year (April start)	`=YEAR(A2)+(MONTH(A2)>=4)`	FY2027
Fiscal quarter (April start)	`=INT(MOD(MONTH(A2)-4,12)/3)+1`	Q3
Start of week (Monday)	`=A2-WEEKDAY(A2,2)+1`	12-Oct-2026
Is weekend?	`=WEEKDAY(A2,2)>5`	FALSE

5. Building a date (calendar) table

A dedicated date table is the standard foundation for time-based analysis in Excel, Power Pivot and Power BI. In Excel 365 you can generate the dates with one formula and then add helper columns beside it.

```
=SEQUENCE(7,1,DATE(2026,10,11),1)
```

This spills 7 consecutive dates starting at 11-Oct-2026. Below are the helper columns for that range, shown in Table 8.

Table 8. A small date table with helper columns

Date	Year	Month	Month name	Qtr	Weekday	Week start (Mon)
11-Oct-2026	2026	10	October	Q4	Sun	05-Oct-2026
12-Oct-2026	2026	10	October	Q4	Mon	12-Oct-2026
13-Oct-2026	2026	10	October	Q4	Tue	12-Oct-2026
14-Oct-2026	2026	10	October	Q4	Wed	12-Oct-2026
15-Oct-2026	2026	10	October	Q4	Thu	12-Oct-2026
16-Oct-2026	2026	10	October	Q4	Fri	12-Oct-2026
17-Oct-2026	2026	10	October	Q4	Sat	12-Oct-2026

Table 9. Formulas behind each helper column (date in A2)

Column	Formula
Year	`=YEAR(A2)`
Month	`=MONTH(A2)`
Month name	`=TEXT(A2,"mmmm")`
Quarter	`="Q"&ROUNDUP(MONTH(A2)/3,0)`
Weekday	`=TEXT(A2,"ddd")`
Week start (Mon)	`=A2-WEEKDAY(A2,2)+1`

> **Note** Notice that 11-Oct-2026 is a Sunday, so its Monday-based week start is 05-Oct-2026, while the other six days share 12-Oct-2026.

6. Worked example: orders, SLA and monthly summary

The dataset below has ten orders. We will add helper columns, flag late shipments against a 5-day service level, and summarise by month. Columns A to E hold the raw data; the date columns are real dates, not text.

Table 10. Raw order data (A = Order ID, B = Order Date, C = Ship Date, D = Region, E = Amount)

Order ID	Order Date	Ship Date	Region	Amount
1001	05-Jan-2026	09-Jan-2026	North	1,200
1002	18-Jan-2026	25-Jan-2026	South	850
1003	03-Feb-2026	05-Feb-2026	East	430
1004	27-Feb-2026	04-Mar-2026	West	1,560
1005	14-Mar-2026	17-Mar-2026	North	920
1006	02-Apr-2026	10-Apr-2026	East	1,780
1007	21-Apr-2026	23-Apr-2026	South	640
1008	09-May-2026	15-May-2026	West	1,100
1009	30-May-2026	03-Jun-2026	North	2,050
1010	12-Jun-2026	16-Jun-2026	East	760

Table 11. Helper columns (F = Order month, G = Quarter, H = Days to ship, I = SLA, J = Weekday)

Order ID	Order month	Quarter	Weekday ordered	Days to ship	SLA (5 days)
1001	Jan-26	Q1	Monday	4	On time
1002	Jan-26	Q1	Sunday	7	Late
1003	Feb-26	Q1	Tuesday	2	On time
1004	Feb-26	Q1	Friday	5	On time
1005	Mar-26	Q1	Saturday	3	On time
1006	Apr-26	Q2	Thursday	8	Late
1007	Apr-26	Q2	Tuesday	2	On time
1008	May-26	Q2	Saturday	6	Late
1009	May-26	Q2	Saturday	4	On time
1010	Jun-26	Q2	Friday	4	On time

Table 12. Helper-column formulas (row 2)

Column	Formula	Purpose
F Order month	`=TEXT(B2,"mmm-yy")`	Label for charts and pivots
G Quarter	`="Q"&ROUNDUP(MONTH(B2)/3,0)`	Quarterly grouping
H Days to ship	`=C2-B2`	Fulfilment time in days
I SLA	`=IF(H2>5,"Late","On time")`	Flags orders above the 5-day target
J Weekday	`=TEXT(B2,"dddd")`	Spot weekday patterns

Monthly summary with date-range criteria

Put the first day of each month in column A of a summary table, then use that date with `EDATE` to build a half-open range: on or after the start, and **before** the start of next month. This pattern also works when your data contains times, because `<=EOMONTH(...)` would miss anything after midnight on the last day.

```
Orders  : =COUNTIFS($B:$B,">="&A2,$B:$B,"<"&EDATE(A2,1))
Total   : =SUMIFS($E:$E,$B:$B,">="&A2,$B:$B,"<"&EDATE(A2,1))
Avg ship: =AVERAGEIFS($H:$H,$B:$B,">="&A2,$B:$B,"<"&EDATE(A2,1))
```

Table 13. Monthly summary (result of the formulas above)

Month start (A)	Orders	Total amount	Avg days to ship
01-Jan-2026	2	2,050	5.5
01-Feb-2026	2	1,990	3.5
01-Mar-2026	1	920	3.0
01-Apr-2026	2	2,420	5.0
01-May-2026	2	3,150	5.0
01-Jun-2026	1	760	4.0
Total	**10**	**11,290**	**4.5**

What the data says

- Q1 (Jan to Mar) sales were **4,960** and Q2 (Apr to Jun) were **6,330**, an increase of 27.6%.
- **3 of 10** orders missed the 5-day target (orders 1002, 1006, 1008). Average days to ship is **4.5**.
- You can count late orders with `=COUNTIF(I:I,"Late")` and the late rate with `=COUNTIF(I:I,"Late")/COUNTA(A2:A11)`.

Grouping dates in a PivotTable

1. Select any cell in the data and choose **Insert > PivotTable**.
2. Drag **Order Date** to Rows and **Amount** to Values.
3. Right-click any date in the pivot, choose **Group**, and select **Months**, **Quarters** and **Years**.
4. Check that Order Date is a true date. Text dates will not group.

> **Tip** Newer versions of Excel auto-group dates when you drag them to Rows. If you do not want that, turn it off in **File > Options > Data**.

7. Cleaning text dates

Data exported from systems, CSV files and websites often arrives as text. A quick check: `=ISNUMBER(A2)` returns TRUE for a real date and FALSE for text. Text dates are usually left-aligned by default. Table 14 shows reliable conversions that do not depend on your regional settings.

Table 14. Converting text to real dates (raw value in A2)

Raw value	Formula	Result
2026-10-15 (text)	`=DATEVALUE(A2)`	15-Oct-2026
15/10/2026 (dd/mm/yyyy text)	`=DATE(RIGHT(A2,4),MID(A2,4,2),LEFT(A2,2))`	15-Oct-2026
20261015 (text or number)	`=DATE(LEFT(A2,4),MID(A2,5,2),RIGHT(A2,2))`	15-Oct-2026
15.10.26 (text)	`=DATE(2000+RIGHT(A2,2),MID(A2,4,2),LEFT(A2,2))`	15-Oct-2026
15-Oct-2026 14:30 (real timestamp)	`=INT(A2)`	15-Oct-2026 (time removed)
15-Oct-2026 14:30 (real timestamp)	`=MOD(A2,1)`	0.604167, shown as 14:30 when formatted as time

Other tools worth knowing:

- **Text to Columns** (Data tab): choose Delimited, skip to step 3, select **Date** and pick the order (DMY, MDY or YMD). Fast for a whole column.
- **Power Query**: change the column type **Using Locale** to read dates correctly regardless of your machine's settings. Best for repeatable monthly imports.
- **Two-digit years**: Excel reads 00 to 29 as 2000 to 2029, and 30 to 99 as 1930 to 1999. Use a 4-digit year whenever possible.

8. Common errors and fixes

Table 15. Troubleshooting guide

Symptom	Likely cause	Fix
A date shows as 46310	Cell format is General or Number	Apply Short Date, or `Ctrl+1` > Date
`#####` in the cell	Column too narrow, or a negative date/time	Widen the column; check the order of subtraction
`#VALUE!` in a date formula	The "date" is actually text	Convert it (Table 14); test with `ISNUMBER`
Day and month swapped (03/04 read as 4 March)	Regional setting differs from the source	Rebuild with `DATE(y,m,d)` or Power Query with locale
`#NUM!` from DATEDIF	Start date is later than end date	Swap the arguments or wrap with `MIN/MAX`
Total time resets after 24 hours	Format is `h:mm`	Use `[h]:mm`
Pivot will not group by month	Blanks or text in the date column	Clean the column; fill or remove blanks
Dates before 1900 fail	Excel does not support them	Store as text or use year/month/day columns

9. Best practices for analysts

- Store dates as **real dates**, one value per cell. Never combine date and text in one cell.
- Prefer **yyyy-mm-dd** when sharing files; it is unambiguous and sorts correctly even as text.
- Build dates with `DATE(y,m,d)` instead of typing strings, so results do not depend on regional settings.
- Use a **half-open range** (>= start, < next start) for date filters. It is safe with timestamps.
- Keep a single **report date** input cell instead of scattering `TODAY()` through the model.
- Use `EDATE`/`EOMONTH` for month logic, not "+30".
- Create a **date table** and relate every fact table to it; it makes fiscal calendars and time intelligence easy.
- Separate **value** from **label**: keep the true date for calculations and use `TEXT()` only for display columns.
- Document your assumptions: weekend pattern, fiscal year start, and whether date ranges include both ends.

10. Practice exercises with answers

Try each task before looking at the formula and answer.

Table 16. Exercises

#	Task	Formula	Answer
1	Days from 1-Jan-2026 to 31-Dec-2026 (end minus start)	`=DATE(2026,12,31)-DATE(2026,1,1)`	364

#	Task	Formula	Answer
2	Last day of February 2028	`=EOMONTH(DATE(2028,2,1),0)`	29-Feb-2028
3	Working days (Mon to Fri) in November 2026	`=NETWORKDAYS(DATE(2026,11,1),DATE(2026,11,30))`	21
4	The date 45 days after 15-Oct-2026	`=DATE(2026,10,15)+45`	29-Nov-2026
5	Fiscal quarter of 15-Aug-2026 (April start)	`=INT(MOD(MONTH(DATE(2026,8,15))-4,12)/3)+1`	Q2
6	Which weekday is 25-Dec-2026? (text)	`=TEXT(DATE(2026,12,25),"dddd")`	Friday

Next step Rebuild Tables 10 to 13 in a blank workbook, then insert a PivotTable and a line chart of monthly totals. Once that works, you have practised every concept in this guide.